

Ternatrees and OEIS A120590

Integral, recurrence, and differential descriptions of an ordered tree class

Bradley Klee, Harm.On.ica S-O-L 5.6 (OpenAI)

Abstract

The sequence $a(n)$ counts rooted ordered ternatrees with n true leaves and no unary branching. Counting trees by their numbers of two-child and three-child vertices gives a finite sum and a contour-integral formula for each coefficient. Summing these integrals gives an integral definition of the ordinary generating function $A(x)$. The local inverse of $\rho(u) = u - 3u^2 - u^3$ then yields the algebraic equation for $A(x)$. Two exact reductions of the same integral family are compared: a shift reduction in n gives a second-order recurrence, while a parameter reduction in x gives a differential equation. Their 6×6 reduction matrices satisfy $G_x = G - x \operatorname{diag}(I_3, 0)$.

Theorem. Let $a(n)$ be the number of rooted ordered ternatrees with n true leaves and no unary branching, and let

$$A(x) = \sum_{n \geq 0} a(n)x^n.$$

Then:

- (1) Put $D(u) = 1 - 3u - u^2$ and $\rho(u) = uD(u)$. For a sufficiently small positively oriented circle γ about the origin,

$$a(0) = 1, \quad a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n} = \frac{1}{n} [u^{n-1}] D(u)^{-n} \quad (n \geq 1).$$

For $|x|$ sufficiently small, with the logarithm chosen continuously from $x = 0$,

$$A(x) = 1 - \frac{1}{2\pi i} \oint_{\gamma} \log\left(1 - \frac{x}{\rho(u)}\right) du.$$

- (2) The integral definition implies

$$4A(x) = 3 + x + A(x)^3, \quad A(0) = 1.$$

- (3) For $n \geq 1$,

$$-3(3n-1)(3n+1)a(n) - 81(n+1)(2n+1)a(n+1) + 13(n+1)(n+2)a(n+2) = 0.$$

- (4) The generating function satisfies

$$(27x^2 + 162x - 13)A''(x) + 27(x+3)A'(x) - 3A(x) = 0.$$

The initial values are

$$1, 1, 3, 19, 150, 1326, 12558, 124590, \dots$$

1 From ternatrees to the integral form

A ternatree is a rooted ordered tree in which every internal vertex has three ordered child positions. A position may be empty, may contain a true leaf, or may contain another internal vertex. Every internal vertex has either two or three occupied positions; unary branching is excluded.

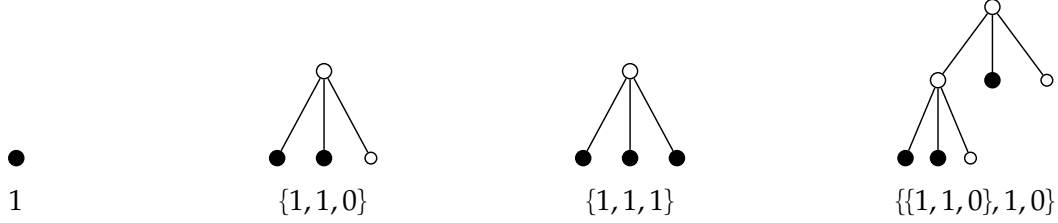


Figure 1. Representative ternatrees. Filled endpoints are true leaves, open endpoints are empty child positions, and horizontal order records the three ordered slots.

Fix a ternatree with n true leaves, i internal vertices having two occupied positions, and j internal vertices having three occupied positions. Counting edges in two ways gives

$$n - 1 = i + 2j.$$

The tree has $m = n + i + j$ vertices. In preorder, write -1 for a leaf, $+1$ for a two-child vertex, and $+2$ for a three-child vertex. The resulting word has total sum -1 . By the cycle lemma, exactly one of its m cyclic rotations has the partial-sum condition of a rooted ordered tree. There are three choices for the empty position at every two-child vertex. Hence the number of trees with this profile is

$$\frac{3^i}{m} \binom{m}{n, i, j}.$$

Putting $j = k$, $i = n - 1 - 2k$, and $m = 2n - 1 - k$ yields

$$a(n) = \sum_{k=0}^{\lfloor (n-1)/2 \rfloor} 3^{n-1-2k} \frac{(2n-2-k)!}{n! (n-1-2k)! k!} \quad (n \geq 1).$$

Equivalently,

$$a(n) = \frac{1}{n} \sum_{k=0}^{\lfloor (n-1)/2 \rfloor} 3^{n-1-2k} \binom{2n-2-k}{n-1} \binom{n-1-k}{k}.$$

Expanding $(1 - 3u - u^2)^{-n}$ gives precisely this sum, so with

$$D(u) = 1 - 3u - u^2,$$

one obtains

$$a(n) = \frac{1}{n} [u^{n-1}] D(u)^{-n}.$$

Cauchy's coefficient formula gives the integral form:

$$\rho(u) = uD(u) = u - 3u^2 - u^3, \quad a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{u^n D(u)^n} = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}.$$

The coefficient integrals may now be summed before any algebraic equation is introduced. Choose γ so that $|x| < |\rho(u)|$ on the contour. Then uniform convergence permits termwise summation:

$$\begin{aligned} A(x) &= 1 + \sum_{n \geq 1} \frac{x^n}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n} \\ &= 1 + \frac{1}{2\pi i} \oint_{\gamma} \sum_{n \geq 1} \frac{1}{n} \left(\frac{x}{\rho(u)} \right)^n du \\ &= 1 - \frac{1}{2\pi i} \oint_{\gamma} \log \left(1 - \frac{x}{\rho(u)} \right) du. \end{aligned}$$

This is the integral definition of the ordinary generating function near the origin.

The algebraic equation follows from this integral definition. Differentiating under the integral sign gives

$$A'(x) = \frac{1}{2\pi i} \oint_{\gamma} \frac{du}{\rho(u) - x}.$$

Since $\rho(0) = 0$ and $\rho'(0) = 1$, there is a unique analytic function $T(x)$ near the origin satisfying $T(0) = 0$ and $\rho(T(x)) = x$. For small x , this is the only zero of $\rho(u) - x$ inside γ . Factor

$$\rho(u) - x = (u - T(x))Q(x, u), \quad Q(x, T(x)) = \rho'(T(x)).$$

Cauchy's integral formula gives

$$A'(x) = \frac{1}{Q(x, T(x))} = \frac{1}{\rho'(T(x))} = T'(x).$$

Because $A(0) = 1$ and $T(0) = 0$, one has $A(x) = 1 + T(x)$. Substitution into $x = \rho(T)$ therefore gives

$$x = (A - 1) - 3(A - 1)^2 - (A - 1)^3,$$

or equivalently

$$4A = 3 + x + A^3.$$

2 Shift reduction and the recurrence

For $n \geq 1$, write

$$H_n(u) = \frac{1}{n\rho(u)^n}, \quad a(n) = \frac{1}{2\pi i} \oint_{\gamma} H_n(u) du.$$

For $w = (w_0, w_1, w_2)^T$, put $\langle w \rangle = w_0 + w_1 u + w_2 u^2$. The fixed reduction operators are

$$U = \frac{1}{13} \begin{pmatrix} 153 & 24 & 3 \\ 72 & 9 & 6 \\ 0 & 0 & 0 \end{pmatrix}, \quad V = \frac{1}{13} \begin{pmatrix} -13 & 0 & 0 \\ 75 & 11 & 3 \\ 24 & 3 & 2 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

They satisfy $\rho \langle Uw \rangle - \rho' \langle Vw \rangle = \langle w \rangle$, while $\langle Jw \rangle = d(\langle w \rangle)/du$. Hence, for $m \geq 1$,

$$\frac{\langle w \rangle}{\rho^{m+1}} = \frac{\langle (U - JV/m)w \rangle}{\rho^m} + \frac{d}{du} \left(\frac{\langle Vw \rangle}{m\rho^m} \right).$$

The integral of the last term around γ is zero, so each application removes one power of ρ and leaves a vector in the same three-dimensional numerator space.

With $e = (1, 0, 0)^T$, define

$$c_0 = e, \quad c_1 = \frac{n}{n+1} \text{Lower}(e, n), \quad c_2 = \frac{n}{n+2} \text{Lower}(\text{Lower}(e, n+1), n).$$

Their third components vanish. The first two components form a 2×3 remainder matrix $X(n)$ whose one-dimensional kernel is generated by

$$P_0(n) = -3(3n-1)(3n+1), \quad P_1(n) = -81(n+1)(2n+1), \quad P_2(n) = 13(n+1)(n+2).$$

The kernel relation lifts to the exact identity

$$P_0 H_n + P_1 H_{n+1} + P_2 H_{n+2} = \frac{d}{du} (R(n, u) H_n(u)),$$

where

$$R(n, u) = \frac{-9nu^4 - 36nu^3 + 6nu^2 + 84nu - 13n - 3u^4 - 12u^3 - 6u^2 + 3u}{\rho(u)}.$$

Integrating around γ gives

$$P_0(n)a(n) + P_1(n)a(n+1) + P_2(n)a(n+2) = 0.$$

3 Direct reduction in the generating variable

Both reductions arise from the coefficient map

$$M_q(a, b) = q(u)a(u) - \rho'(u)b(u),$$

where $a(u)$ and $b(u)$ have degree less than three. In the basis $1, u, \dots, u^5$, the specialization $q = \rho$ is represented by

$$G = \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 6 & -1 & 0 \\ -3 & 1 & 0 & 3 & 6 & -1 \\ -1 & -3 & 1 & 0 & 3 & 6 \\ 0 & -1 & -3 & 0 & 0 & 3 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}, \quad \det G = -13.$$

Let $E = \begin{pmatrix} I_3 \\ 0 \end{pmatrix}$. The relation

$$G^{-1}E = \begin{pmatrix} U \\ V \end{pmatrix}$$

produces exactly the operators used in Section 2. Thus G is the fixed coefficient map for the shift reduction and the natural starting point for the direct comparison.

Differentiating the generating-function integral obtained in Section 1 gives

$$A'(x) = \frac{1}{2\pi i} \oint_{\gamma} F(x, u) du, \quad F(x, u) = \frac{1}{g(x, u)}, \quad g(x, u) = \rho(u) - x.$$

Replacing $q = \rho$ by $q = g$ changes only the upper-left block of the coefficient map:

$$G_x = \begin{pmatrix} -x & 0 & 0 & -1 & 0 & 0 \\ 1 & -x & 0 & 6 & -1 & 0 \\ -3 & 1 & -x & 3 & 6 & -1 \\ -1 & -3 & 1 & 0 & 3 & 6 \\ 0 & -1 & -3 & 0 & 0 & 3 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}.$$

Consequently,

$$G_x = G - x \begin{pmatrix} I_3 & 0 \\ 0 & 0 \end{pmatrix} = G - xEE^T, \quad \det G_x = 27x^2 + 162x - 13, \quad G_x^{-1}E = \begin{pmatrix} U_x \\ V_x \end{pmatrix}.$$

The same pole-lowering identity holds with ρ replaced by g and (U, V) replaced by (U_x, V_x) . Reducing F , $\partial_x F$, and $\partial_x^2 F$ gives three columns in the same two-dimensional remainder space. Their kernel is generated by

$$\begin{pmatrix} 24 \\ 81x + 243 \\ 27x^2 + 162x - 13 \end{pmatrix}.$$

The corresponding exact identity is

$$24F + (81x + 243)\partial_x F + (27x^2 + 162x - 13)\partial_x^2 F = \frac{d}{du} \left(\frac{N(x, u)}{g(x, u)^2} \right),$$

where

$$N(x, u) = 12u^4 + 48u^3 + 3ux - 87u + 3x + 13.$$

After integration around γ ,

$$(27x^2 + 162x - 13)A'''(x) + (81x + 243)A''(x) + 24A'(x) = 0.$$

This is the derivative of

$$(27x^2 + 162x - 13)A''(x) + 27(x + 3)A'(x) - 3A(x) = C.$$

The initial values $A(0) = 1$, $A'(0) = 1$, and $A''(0) = 6$ give $C = 0$.

4 Comparison of the two reductions

The two calculations are instances of the same exact reduction. They differ only in the family placed under the contour integral and in the coefficient field of the inverse matrix.

	Shift in n	Derivative in x
Integral family	$H_{n+r}(u) = 1/((n+r)\rho(u)^{n+r})$	$\partial_x^r F(x, u) = r!/g(x, u)^{r+1}$
Denominator	$\rho(u)$	$g(x, u) = \rho(u) - x$
Coefficient map	G	$G_x = G - xEE^T$
Inverse blocks	U, V over $\mathbb{Q}$	U_x, V_x over $\mathbb{Q}(x)$
Reduced columns	H_n, H_{n+1}, H_{n+2}	$F, \partial_x F, \partial_x^2 F$
Kernel	$(P_0(n), P_1(n), P_2(n))^T$	$(24, 81x + 243, \det G_x)^T$
Integral consequence	recurrence for $a(n)$	differential equation for $A(x)$

At $x = 0$, the direct matrix specializes to the shift matrix:

$$G_x|_{x=0} = G, \quad \det G_x|_{x=0} = \det G = -13.$$

The determinant plays the same structural role in both calculations. Its constant value supplies the denominator 13 in U and V and the leading coefficient $13(n+1)(n+2)$ of the recurrence. Its polynomial deformation is the leading coefficient of the differential equation.

The differential equation reproduces the recurrence without any further symbolic reduction. Multiplying it by -1 gives

$$(13 - 162x - 27x^2)A'' - 27(x + 3)A' + 3A = 0.$$

The coefficient of x^n is

$$(-27n^2 + 3)a(n) + (-162n^2 - 243n - 81)a(n+1) + (13n^2 + 39n + 26)a(n+2),$$

which is exactly

$$P_0(n)a(n) + P_1(n)a(n+1) + P_2(n)a(n+2).$$

Thus the two kernels encode the same relation in the discrete and generating variables.

The shared mechanism may be summarized by writing

$$M_q(a, b) = q(u)a(u) - \rho'(u)b(u), \quad M_\rho = G, \quad M_g = G_x.$$

If $M_q^{-1}E = \binom{U_q}{V_q}$, then both calculations use

$$\frac{\langle w \rangle}{q^{m+1}} = \frac{\langle (U_q - JV_q/m)w \rangle}{q^m} + \frac{d}{du} \left(\frac{\langle V_q w \rangle}{mq^m} \right).$$

The inverses display the parallel leading factors directly:

$$G^{-1} = \frac{\text{adj}(G)}{-13}, \quad G_x^{-1} = \frac{\text{adj}(G_x)}{27x^2 + 162x - 13}.$$

In each route, three reduced columns lie in a two-dimensional space:

$$\begin{aligned} (H_n, H_{n+1}, H_{n+2}) &\longmapsto X(n) \longmapsto \ker X(n) = \langle P(n) \rangle, \\ (F, \partial_x F, \partial_x^2 F) &\longmapsto X_x(x) \longmapsto \ker X_x(x) = \langle C(x) \rangle. \end{aligned}$$

5 Reference pseudocode

The following specification implements the shift reduction. It constructs the remainder matrix, extracts its primitive kernel vector, generates the sequence, and converts the recurrence into the differential equation.

```

(Target:  $4^*A(x)=3+x+A(x)^3$ ,  $A(0)=1$ ; print  $a(0)..a(N)$ .)
(n,x,A,A',A'' are formal; k,N are integers; arithmetic is exact.)
(Matrices are 1-based; vectors are columns; ranges include both ends.)
(ExactQuotient asserts zero integer remainder; Factor only prints.)
Function Lower(U,V,J,w,m)
1: b <- V*w;
2: Return U*w-(J*b)/m;           (run the quotient rule backward)
Function Normalize3(p0,p1,p2)
1: d <- LCM(Denom(Together(p0)),Denom(Together(p1)),Denom(Together(p2)));
2: {q0,q1,q2} <- Expand(d*{p0,p1,p2});
3: h <- PolynomialGCD(q0,q1,q2); {q0,q1,q2} <- {q0/h,q1/h,q2/h};
4: c <- LCM of all coefficient denominators in q0,q1,q2;
5: {q0,q1,q2} <- Expand(c*{q0,q1,q2});
6: g <- GCD of all integer coefficients; {q0,q1,q2} <- {q0/g,q1/g,q2/g};
7: If LeadingCoefficient(q2,n)<0, negate all three;
8: Return Factor({q0,q1,q2});
Function Cancel3(X)              (X has two rows, three columns)
1: p0 <- X[1,2]*X[2,3]-X[1,3]*X[2,2]; (cancel columns 2 and 3)
2: p1 <- X[1,3]*X[2,1]-X[1,1]*X[2,3]; (cancel columns 1 and 3)
3: p2 <- X[1,1]*X[2,2]-X[1,2]*X[2,1]; (cancel columns 1 and 2)
4: Return Normalize3(p0,p1,p2);
Function Apply2(P,r)
1: Write P(n)=a*n^2+b*n+c;
2: Return a*(x^2*A''+(1-2*r)*x*A'+r^2*A)+b*(x*A'-r*A)+c*A;
Function MakeODE(P0,P1,P2)
1: L <- x^2*Apply2(P0,0)+x*Apply2(P1,1)+Apply2(P2,2);
2: L <- Collect(Expand(L/x^2),{A'',A',A}); (division is exact)
3: Factor each polynomial multiplying A'', A', and A;
4: If the coefficient of A'' leads negatively in x, set L <- -L;
5: Return L;
Function Q3(N)
1: Require N>=2;
2: U <- (1/13)*[[153,24,3],[72,9,6],[0,0,0]];
3: V <- (1/13)*[[-13,0,0],[75,11,3],[24,3,2]];
4: J <- [[0,1,0],[0,0,2],[0,0,0]]; e <- [1,0,0]^T;
5: c0 <- e; c1 <- (n/(n+1))*Lower(U,V,J,e,n);
6: c2 <- (n/(n+2))*Lower(U,V,J,Lower(U,V,J,e,n+1),n);
7: X <- first two rows of MatrixFromColumns(c0,c1,c2);
8: {P0,P1,P2} <- Cancel3(X); Assert Simplify(X*{P0,P1,P2}^T)=0;
9: a[0] <- 1; a[1] <- ExactQuotient(1,4-3*a[0]^2);
10: a[2] <- ExactQuotient(3*a[0]*a[1]^2,4-3*a[0]^2);
11: For k <- 1 to N-2:
12:   a[k+2] <- ExactQuotient(-P0(k)*a[k]-P1(k)*a[k+1],P2(k));
13: S <- Sum(a[k]*x^k,k=0..N); Assert Coeffs(4*S-3-x-S^3,0..N)=0;
14: ODE <- MakeODE(P0,P1,P2);
15: Assert Coeffs(Substitute(ODE,A=S,A'=S',A''=S''),0..N-2)=0;
16: Return {a[0..N],Factor({P0,P1,P2}),ODE=0};

```

6 Pseudocode operations and exact realization

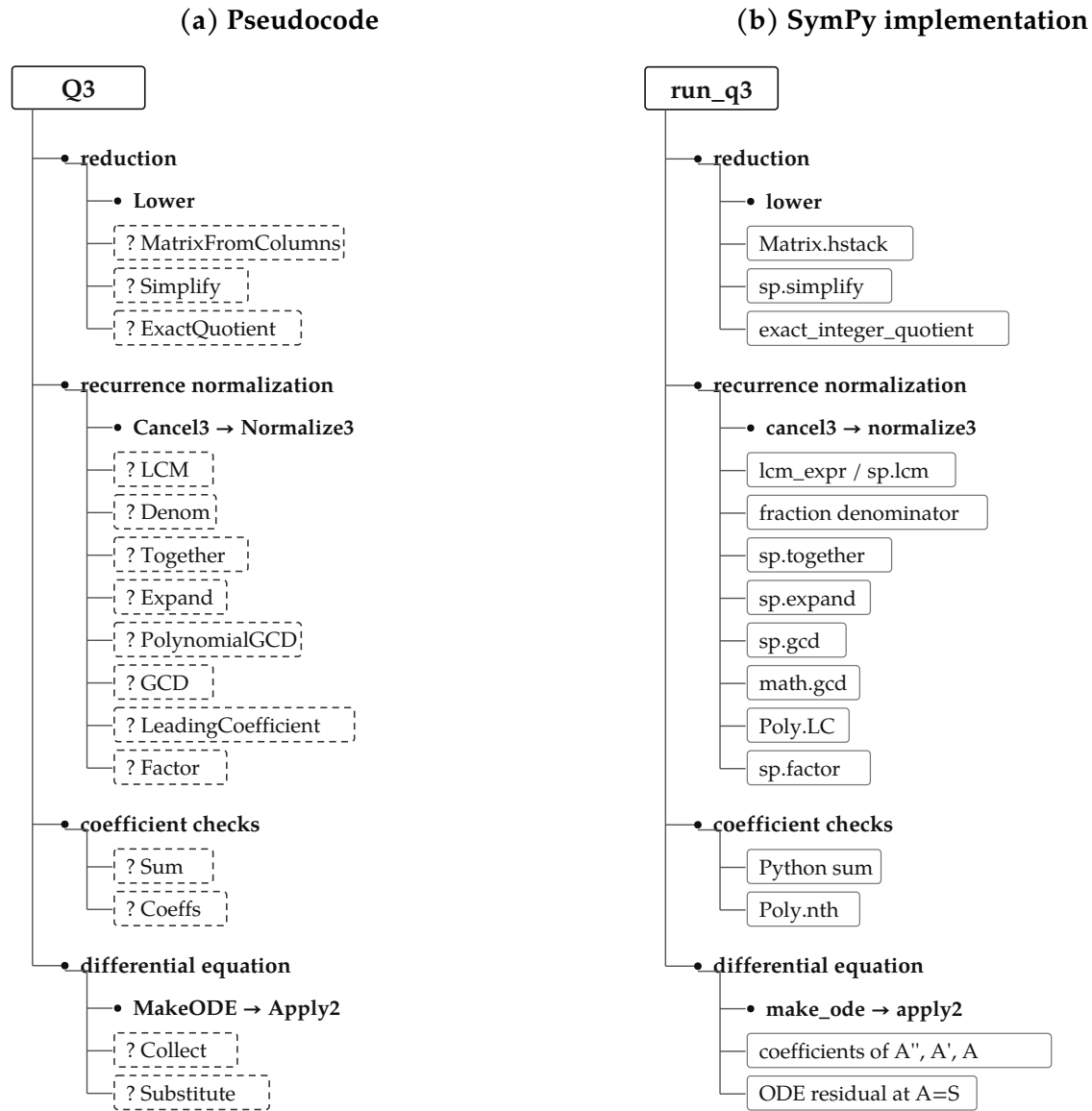


Figure 2. The local routines and their algebraic dependencies. Dashed leaves are operations used but not defined in the pseudocode; the corresponding rows on the right identify their exact realization.

7 Relations among the formulas

The geometric count, integral definition, recurrence, algebraic equation, and differential equation determine the same series. Their relations are

$$\text{ternatree profiles} \longrightarrow a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n} \longrightarrow A(x) = 1 - \frac{1}{2\pi i} \oint_{\gamma} \log\left(1 - \frac{x}{\rho(u)}\right) du,$$

$$\text{local inverse } \rho(T) = x \longrightarrow A = 1 + T \longrightarrow 4A = 3 + x + A^3,$$

$$\text{shift reduction of the coefficient integral} \longrightarrow \text{recurrence},$$

$$\text{parameter reduction of the generating-function integral} \longrightarrow \text{differential equation}.$$

The recurrence and differential equation correspond coefficient by coefficient, and either one generates the terms of A120590 from the stated initial values.

A Exact SymPy implementation

This implementation merges one-shot code produced independently by Kimi (Moonshot AI) and Claude (Anthropic AI), both working from the pseudocode presented in Section 5.

A.1 Arithmetic and one-step lowering

The functions below define exact least common multiples and the map $w \mapsto (U - JV/m)w$.

```

1 #!/usr/bin/env python3
2 """Derive the A120590 recurrence, terms, and ODE from fixed q=3 data."""
3
4 from __future__ import annotations
5
6 import argparse
7 import math
8 from typing import Iterable
9
10 import sympy as sp
11
12
13 def lcm_expr(values: Iterable[sp.Expr]) -> sp.Expr:
14     result = sp.Integer(1)
15     for value in values:
16         result = sp.lcm(result, sp.simplify(value))
17     return result
18
19
20 def lower(U: sp.Matrix, V: sp.Matrix, J: sp.Matrix,
21          w: sp.Matrix, m: sp.Expr) -> sp.Matrix:
22     b = V * w
23     return sp.simplify(U * w - (J * b) / m)

```

A.2 Primitive recurrence normalization

This function clears rational denominators, removes common polynomial and integer factors, and fixes the sign of the recurrence vector.

```

26 def normalize3(p0: sp.Expr, p1: sp.Expr, p2: sp.Expr,
27               n: sp.Symbol) -> tuple[sp.Expr, sp.Expr, sp.Expr]:
28     values = [sp.together(p0), sp.together(p1), sp.together(p2)]
29     denoms = [sp.fraction(value)[1] for value in values]
30     common = lcm_expr(denoms)
31     q = [sp.expand(sp.cancel(common * value)) for value in values]
32
33     poly_gcd = sp.gcd(sp.gcd(q[0], q[1]), q[2])
34     if poly_gcd != 0:
35         q = [sp.expand(sp.cancel(value / poly_gcd)) for value in q]
36
37     coeff_denoms: List[sp.Expr] = []
38     for value in q:
39         coeff_denoms.extend(
40             sp.fraction(c)[1] for c in sp.Poly(value, n).all_coeffs()
41         )
42     common_coeff = lcm_expr(coeff_denoms)
43     q = [sp.expand(common_coeff * value) for value in q]
44
45     integer_coeffs: List[int] = []
46     for value in q:
47         integer_coeffs.extend(
48             abs(int(c)) for c in sp.Poly(value, n).all_coeffs() if c != 0
49         )
50     content = 0
51     for coeff in integer_coeffs:
52         content = math.gcd(content, coeff)
53     if content > 1:
54         q = [sp.expand(value / content) for value in q]
55
56     if sp.Poly(q[2], n).LC() < 0:
57         q = [-value for value in q]
58     return tuple(sp.factor(value) for value in q)

```

A.3 Kernel vector and recurrence-to-ODE map

The 2×3 minors form the kernel vector. The remaining functions convert quadratic recurrence coefficients into the coefficients of A'' , A' , and A .

```

61 def cancel3(X: sp.Matrix, n: sp.Symbol
62             ) -> tuple[sp.Expr, sp.Expr, sp.Expr]:
63     p0 = X[0, 1] * X[1, 2] - X[0, 2] * X[1, 1]
64     p1 = X[0, 2] * X[1, 0] - X[0, 0] * X[1, 2]
65     p2 = X[0, 0] * X[1, 1] - X[0, 1] * X[1, 0]
66     return normalize3(p0, p1, p2, n)
67
68
69 def apply2(P: sp.Expr, r: int, n: sp.Symbol, x: sp.Symbol,
70           A: sp.Symbol, Ap: sp.Symbol, App: sp.Symbol) -> sp.Expr:
71     poly = sp.Poly(sp.expand(P), n)
72     a = poly.coeff_monomial(n**2)
73     b = poly.coeff_monomial(n)
74     c = poly.coeff_monomial(1)
75     return (
76         a * (x**2 * App + (1 - 2 * r) * x * Ap + r**2 * A)
77         + b * (x * Ap - r * A)
78         + c * A
79     )
80
81
82 def make_ode(P0: sp.Expr, P1: sp.Expr, P2: sp.Expr,
83             n: sp.Symbol, x: sp.Symbol
84             ) -> tuple[sp.Expr, sp.Expr, sp.Expr]:
85     A, Ap, App = sp.symbols("A Ap App")
86     L = (
87         x**2 * apply2(P0, 0, n, x, A, Ap, App)
88         + x * apply2(P1, 1, n, x, A, Ap, App)
89         + apply2(P2, 2, n, x, A, Ap, App)
90     )
91     L = sp.expand(sp.cancel(L / x**2))
92     coeffs = (
93         sp.factor(L.coeff(App)),
94         sp.factor(L.coeff(Ap)),
95         sp.factor(L.coeff(A)),
96     )
97     if sp.Poly(coeffs[0], x).LC() < 0:
98         coeffs = tuple(-value for value in coeffs)
99     return coeffs

```

A.4 Term generation and exact checks

The driver enters U, V, J , constructs X , derives the recurrence, generates terms, and checks the algebraic and differential equations.

```

102 def exact_integer_quotient(number: sp.Expr, denom: sp.Expr) -> int:
103     value = sp.cancel(sp.simplify(number) / sp.simplify(denom))
104     if value.is_Integer is not True:
105         raise ArithmeticError(f"nonintegral quotient: ({number})/({denom})")
106     return int(value)
107
108
109 def run_q3(N: int) -> dict[str, object]:
110     if N < 2:
111         raise ValueError("N must be at least 2")
112
113     n, x = sp.symbols("n x")
114     U = sp.Rational(1, 13) * sp.Matrix(
115         [[153, 24, 3], [72, 9, 6], [0, 0, 0]]
116     )
117     V = sp.Rational(1, 13) * sp.Matrix(
118         [[-13, 0, 0], [75, 11, 3], [24, 3, 2]]
119     )
120     J = sp.Matrix([[0, 1, 0], [0, 0, 2], [0, 0, 0]])
121     e = sp.Matrix([1, 0, 0])
122
123     c0 = e
124     c1 = sp.simplify((n / (n + 1)) * lower(U, V, J, e, n))
125     c2 = sp.simplify(
126         (n / (n + 2)) * lower(U, V, J, lower(U, V, J, e, n + 1), n)
127     )
128     X = sp.simplify(sp.Matrix.hstack(c0, c1, c2)[:2, :])
129     P0, P1, P2 = cancel3(X, n)
130
131     if sp.simplify(X * sp.Matrix([P0, P1, P2])) != sp.zeros(2, 1):
132         raise AssertionError("X*P is not zero")
133
134     terms = [0] * (N + 1)
135     terms[0] = 1
136     terms[1] = exact_integer_quotient(1, 4 - 3 * terms[0] ** 2)
137     terms[2] = exact_integer_quotient(
138         3 * terms[0] * terms[1] ** 2,
139         4 - 3 * terms[0] ** 2,
140     )
141     for k in range(1, N - 1):
142         numer = -P0.subs(n, k) * terms[k] - P1.subs(n, k) * terms[k + 1]
143         denom = P2.subs(n, k)
144         terms[k + 2] = exact_integer_quotient(numer, denom)
145
146     S = sum(sp.Integer(terms[k]) * x**k for k in range(N + 1))
147     cubic = sp.Poly(sp.expand(4 * S - 3 * x - S**3), x)
148     if any(cubic.nth(k) != 0 for k in range(N + 1)):
149         raise AssertionError("truncated series fails the cubic check")
150
151     ode = make_ode(P0, P1, P2, n, x)
152     residual = sp.Poly(
153         sp.expand(ode[0] * sp.diff(S, x, 2)
154             + ode[1] * sp.diff(S, x) + ode[2] * S),
155         x,
156     )
157     if any(residual.nth(k) != 0 for k in range(N - 1)):
158         raise AssertionError("truncated series fails the ODE check")
159
160     return {
161         "n": n,
162         "x": x,
163         "X": X,
164         "P": (P0, P1, P2),
165         "terms": terms,
166         "ode": ode,
167     }

```